

EXTENDS *Naturals, Sequences*

$$\begin{aligned} Keys &\triangleq \{ \text{"checking"}, \text{"savings"} \} \\ Values &\triangleq \{0, 1, 2\} \\ InitVal &\triangleq 0 \end{aligned}$$
We define the *Workload* as a Sequence (Tuple) of Records.
$$\begin{aligned} Workload &\triangleq \langle \text{ T1: TransactSavings (PSI)} \\ &\quad [\\ &\quad \quad session \mapsto \text{"client_1"}, \\ &\quad \quad level \mapsto \text{"PSI"}, \\ &\quad \quad ops \mapsto \langle [type \mapsto \text{"R"}, key \mapsto \text{"checking"}], \\ &\quad \quad \quad [type \mapsto \text{"R"}, key \mapsto \text{"savings"}], [type \mapsto \text{"W"}, key \mapsto \text{"savings"}, val \mapsto 1] \rangle \\ &\quad], \text{ T2: WriteCheck (SER)} \\ &\quad [\\ &\quad \quad session \mapsto \text{"client_2"}, \\ &\quad \quad level \mapsto \text{"SER"}, \\ &\quad \quad ops \mapsto \langle [type \mapsto \text{"R"}, key \mapsto \text{"checking"}], \\ &\quad \quad \quad [type \mapsto \text{"R"}, key \mapsto \text{"savings"}], [type \mapsto \text{"W"}, key \mapsto \text{"checking"}, val \mapsto 2] \rangle \\ &\quad] \\ &\rangle \end{aligned}$$

VARIABLES

$$\begin{aligned} &client_dep, \\ &client_last_commit, \\ &router_ct, \\ &shard_ct, \\ &tx_status, \\ &tx_snapshot, \\ &tx_checkset, \\ &tx_buffer, \\ &tx_commit_time, \\ &pc, \\ &store, \\ &op_seq \end{aligned}$$

$$\begin{aligned} MI &\triangleq \text{INSTANCE } MixedIsolation \text{ WITH} \\ &\quad Keys \leftarrow Keys, \\ &\quad Values \leftarrow Values, \\ &\quad InitVal \leftarrow InitVal, \\ &\quad Workload \leftarrow Workload \end{aligned}$$

$$\begin{aligned} Spec &\triangleq MI!Spec \\ OperationalSafety &\triangleq MI!OperationalSafety \end{aligned}$$